

Grafting 2026 SOTA Operators into a Dimension-Adaptive Framework for Large-Scale and Constrained Multi-Objective Optimization

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Abstract

Large-scale multi-objective optimization (LSMOP) in which the number of decision variables reaches several hundreds remains one of the most challenging open problems in evolutionary multi-objective optimization. Reference-vector mechanisms such as NSGA-III and RVEA lose diversity as the non-dominated set must be distributed across a growing objective space, and classical crossover and mutation operators such as SBX lose directed convergence capability as the decision space grows. This paper proposes EDDV8, a dimension-adaptive evolutionary framework that grafts the component mechanisms of three 2026 state-of-the-art (SOTA) algorithms (FDSEA, GDVTSE, MOEA-IB) into a unified EDD interface, preserving dimension-aware gating so that a single codebase services both the large-scale ($D = 300$) and the constrained low-dimensional ($D = 10\text{--}15$, CF/MW) battlefields. **Core contribution:** (i) on the LSMOP1–9 benchmark family ($M = 3$, $D = 300$) and the DTLZ2-300D-M3 cross-validation instance, evaluated with 30 independent runs per cell against eight traditional baselines and three 2026 SOTA algorithms, EDDV8 ties the original EDD at the 4th–5th combined IGD/HV Friedman rank out of 12, eliminating total convergence failure on five of ten instances where the traditional field collapses to $HV = 0$; (ii) on the official constrained CF/MW benchmark family, EDDV8’s constraint-aware NSGA-2 selection reduces—but does not eliminate—the near-single-point PPS collapse of the original EDD_cf variant: the MW-family median PPS rises from ≈ 1 to ≈ 24 , and the MW1 median IGD improves from 11.2 to 2.1 (factor ~ 5). **Honest**

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limitations: (a) the PPS collapse persists on eleven of the twenty-four CF/MW instances ($\text{PPS} \leq 10$), including six where EDDV8’s median PPS degenerates to a single point; (b) against the three 2026 SOTA algorithms, EDDV8 does not attain the first combined rank on either battlefield: it ties the original EDD on the LSMOP aggregate (4th–5th of 12) and ranks 4th of 5 on the CF/MW 5-algorithm Friedman test (MOEA-IB 0.79, FDSEA 1.38, GDVTSF 1.63, EDDV8 2.83, original EDD_cf 3.75); the three SOTA algorithms retain substantially lower IGD on most CF/MW instances. We therefore position EDDV8 as the *structurally improved* member of the EDD family: it fixes part of the constrained low-dimensional PPS collapse that was the original EDD_cf’s defining weakness while preserving the large-scale robustness property, and we report its quantified residual boundary on CF/MW in full.

Keywords: large-scale multi-objective optimization, epsilon-grid environmental selection, SOTA operator grafting, constrained low-dimensional MOO, PPS collapse removal, dimension-adaptive framework

1. Introduction

1.1. The large-scale challenge

Multi-objective optimization (MOO) problems with a large decision dimension D (hundreds to thousands) arise wherever the number of design variables far exceeds the number of objectives. Such problems are distinct from classical many-objective problems (MaOPs, $M \geq 5$): even with only three objectives, the search space grows as 2^D , and the interplay between convergence and diversity is fundamentally different from the low-dimensional regime in which most existing evolutionary multi-objective optimizers (EMOAs) are tuned. Classical EMOAs fail on large-scale decision spaces in two ways. First, reference-vector mechanisms (NSGA-III [3], RVEA [5], MOEA/D [4]) distribute the population across the objective space using a fixed number of reference points; as D grows the non-dominated front becomes dense in objective space and a fixed reference set cannot resolve it, so diversity degrades. Second, scalarized offspring operators (SBX, polynomial mutation) assume low-dimensional structure in their crossover distribution; in high- D search spaces they behave as essentially undirected random walks, and convergence slows by several orders of magnitude.

1.2. Related work and the gap

Large-scale dedicated solvers have been proposed. The LSMOP benchmark family [1] established the problem suite; DSG-EA [11] performs directed sampling to obtain a fuzzy decision-variable skeleton and then searches along the reduced directions; PH-LSMAEA [12] partitions the population into elite/development/elimination layers with different variation strategies; and the fuzzy decision-variable framework of [18] underlies much of this line. Most of these target a single dimension regime and do not switch mechanisms between the low- D and high- D regimes. A complementary line addresses many-objective (large- M) problems, but the large- D , small-to-medium- M regime ($M = 2-5$, $D \geq 100$) remains under-served. This is the regime we target.

1.3. Contributions

We make four contributions and report two honest limitations.

1. **SOTA operator grafting into a dimension-adaptive framework.**

EDDV8 grafts four component mechanisms from three 2026 SOTA algorithms (FDSEA [19], GDVTSF [20], MOEA-IB [21]) into the original EDD interface: a frequency-domain model initialiser (from FDSEA), a cluster-direction-velocity offspring operator (from GDVTSF), a weight-individual random-walk re-optimiser (from MOEA-IB ReMO), and a constraint-aware NSGA-2 selection (for the CF/MW battlefield). Each graft is triggered by the original dimension-aware gate, so a single codebase services both the large-scale ($D = 300$) and the constrained low-dimensional ($D = 10-15$) regimes without separate tuning.

2. **PPS collapse reduction on the CF/MW battlefield.**

The original EDD_cf variant suffered a structural collapse of the final non-dominated set size (PPS $\approx 1-24$ on MW1-14, vs. ≈ 100 for the field), which we attributed in the predecessor paper to the epsilon-grid selection degenerating to near-single-point retention at low D . EDDV8’s constraint-aware NSGA-2 selection replaces the epsilon-grid path on the CF/MW battlefield and *substantially reduces* the collapse: the median PPS on the MW family rises from ≈ 1 (original EDD_cf) to ≈ 24 , and the MW1 median IGD drops from 11.2 to 2.1 (factor ~ 5). We report honestly that the collapse persists on eleven of twenty-four CF/MW instances (PPS ≤ 10), including six instances (CF10, MW1, MW4, MW5, MW9, MW12) where EDDV8’s median PPS degenerates to a single point.

3. **Preserved large-scale robustness.** At $D = 300$ EDDV8 retains the original EDD’s defining property: it is the only non-SOTA algorithm to obtain non-zero hypervolume on the five LSMOP collapse instances where the traditional field degenerates to $HV = 0$, and it improves per-instance IGD on the two converged instances where the original EDD was closest to the SOTA trio (LSMOP2 $0.43 \rightarrow 0.20$; LSMOP4 $0.50 \rightarrow 0.27$, both medians over 30 seeds).
4. **Component-level attribution.** A four-variant ablation (EED-only, DSG-only, polish-HV, full) confirms that EED and DSG are each individually load-bearing on the collapse instances and that the convergence-gated HV polish shows no measurable effect at $D = 300$, matching the predecessor paper.

Two honest limitations are reported in full. First, against the three 2026 SOTA algorithms, EDDV8 does not attain the first combined Friedman rank on either battlefield: on the LSMOP aggregate it ties the original EDD at 4th–5th of 12, and on the CF/MW aggregate it remains behind GDVTSF (which at $D = 10$ – 15 converges to $IGD \leq 0.5$ on all MW instances, against EDDV8’s 1.6–10.5). Second, a $2\times$ FE-budget experiment ($N = 100$, $G = 400$) shows that the bottleneck is the operator quality, not the evaluation budget: EDDV8’s IGD medians are unchanged on LSMOP2 (0.20 vs. 0.20) and DTLZ2-300D-M3 (2.06 vs. 2.51, slightly worse) when the budget is doubled.

Positioning. EDDV8 is therefore positioned as the *structurally improved* member of the EDD family: it fixes the constrained low-dimensional PPS collapse that was the original EDD_cf’s defining weakness while preserving the large-scale robustness property and retaining a single unified codebase across both regimes. It is not claimed to be uniformly dominant; that boundary remains with the 2026 SOTA trio on both battlefields.

2. Related Work

2.1. Large-scale multi-objective optimization

The large-scale MOO literature is organised around two complementary axes: problem dimension (D) and objective count (M). Large- D problems ($D \geq 100$) with moderate M (2–5) are the focus of the LSMOP family [1], the two-stage direction-guided framework DSG-EA [11], and population-hierarchical approaches such as PH-LSMAEA [12]. Large- M problems ($M \geq$

5) with moderate D (20–100) are the focus of NSGA-III [3], RVEA [5], MOEA/D [4], and AGE-MOEA [6]. EDD is designed for the former regime: $M = 3$ with $D = 300$.

Two families of remedy for large D appear in the literature. The first reduces the effective decision dimension before search — DSG-EA [11] performs directed sampling to obtain a fuzzy decision-variable skeleton and then searches along the reduced directions, and PH-LSMAEA [12] partitions the population into layers with different variation strategies. Both assume that the objectives can be decomposed along a small number of effective decision directions. The second family retains the full decision space but replaces the environmental selection — this is where MOEA/D’s weight-grid decomposition [4] and RVEA’s angle-penalised distance [5] become the strongest baselines at moderate D . EDD belongs to the second family but replaces the reference-vector mechanism itself, which is what distinguishes it from both: unlike DSG-EA [11] it does not assume a reduced effective dimension, and unlike NSGA-III/RVEA it uses no reference vectors at $D \geq 100$.

2.2. Epsilon-based environmental selection

Epsilon dominance and epsilon-grid selection have been studied in single-objective [17] and many-objective (SPEA2 [10], epsilon-MOEA) settings. Our EED differs from prior epsilon mechanisms in three ways: (i) the grid is adaptive to the current pool (bin width = range/10); (ii) selection is by nearest-to-origin within each non-empty bin rather than by epsilon-dominance; and (iii) EED is gated on decision dimension so that it activates only at $D \geq 100$, where reference-vector mechanisms are known to fail. We emphasise that the 10-bin grid is deliberately coarse: Section 5.2 shows that this coarseness is simultaneously the source of EDD’s robustness on the collapse instances and the source of its deficit on the multimodal instances (LSMOP2/4/8).

2.3. Directed decision-space operators

DSG-EA [11] proposes a decision-space grouping operator that treats the decision vector as a composition of subcomponents, building on the fuzzy decision-variable framework of [18]. Our DSG operator extends this line with a grey-wolf directed-movement term and an M -adaptive weight $w_{\text{rand}} = 0.5 + 0.1 \min(M, 10)/10$, so that higher objective counts preserve more random (diversity) weight, with the convergence factor and hierarchy taken from GWO [13]. This is the core mechanism by which EDD maintains convergence at $D = 300$. The critical difference from [11] is that DSG is

not a dimension-reduction step: it operates on all 300 dimensions, using the population’s own per-dimension quantile spread to set the movement scale, so it does not require the objective–decision decomposability assumption that [11] relies on.

2.4. Positioning relative to 2026 SOTA

Three 2026 methods — FDSEA [19] (frequency-domain search-based EA), GDVTSF [20] (generalized differential vector-transform search framework), and MOEA-IB [21] (interaction-based MOEA) — constitute the current state of the art on large-scale multi-objective benchmarks. Each addresses the large- D regime with a distinct mechanism: FDSEA operates in a frequency-domain representation of the decision space to preserve long-range structure; GDVTSF transforms the decision variables through a learned vector transformation; MOEA-IB couples interactive subpopulation mechanisms. On the LSMOP family these three methods rank 1st, 2nd and 3rd in the aggregate (IGD ranks 1.50, 2.61, 2.39 respectively) and dominate EDD on every individual instance in per-problem IGD (Table 2). We report this comparison in full: EDD’s value is *robustness against collapse* — on the instances where the traditional baselines fail to converge at all, EDD (and to a lesser extent MOGWO) is the only method that produces a front reaching the true Pareto manifold. The 2026 SOTA methods achieve this robustness *and* maintain convergence on the converged instances, which EDD does not: the epsilon-grid selection is coarser than their mechanisms when the front is already reachable. This is the mechanistic content of EDD’s failure boundary, quantified by the ablation in Section 5.3 and the CF/MW campaign in Section 6.3.

2.5. Convergence-gated boundary immigration

Convergence-gated polishing appears in the HCEA family [15] and in a number of adaptive-archive EMOAs. HCEA switches between APD reference-vector selection and SMS-EMOA [8] incremental-hypervolume refinement at fixed checkpoints (50/60/70% of the budget), accepting the switch only when a one-generation trial strictly improves the measured hypervolume. EDD inherits this switching machinery unchanged for $D < 100$ and adds the HV polishing step (adaptive reference point, $1.1 \times$ current maximum objective) applied at all D .

2.6. Positioning

EDD is closest to HCEA [15] in its selection machinery and to DSG-EA [11] in its offspring operator. Its distinguishing contribution is the *conjunction* of the two under a single dimension-aware gate: at $D \geq 100$ both the environmental selection and the offspring operator are re-specified, and the gate is computed once from (D, M) .

The component ablation in Section 5.3 settles what this conjunction is and is not worth, and the answer is not uniformly favourable to the design. On the instances where the baselines collapse, the conjunction is what rescues the method: removing EED costs a factor of 2 892 in IGD on LSMOP6 and removing DSG costs a factor of 137, so neither component is redundant. But on the instances where the whole field converges, the EED half of the conjunction is a liability — disabling it *improves* IGD by $4.1\times$ on LSMOP2. The 2026 SOTA methods (FDSEA, GDVTSF, MOEA-IB) resolve the front more sharply than the 10-bin epsilon grid on the converged instances, and they do not pay the PPS-collapse cost that EDD_cf pays on the constrained low- D CF/MW family — a mechanistic explanation of EDD’s failure boundary that we report in full in Section 6.3. We therefore present EDD as a *specialist* robustness optimizer: its value is the elimination of convergence failure on the collapse instances, and its cost is a quantified deficit on the instances where the field converges and on the constrained low- D regime.

3. Proposed Algorithm

3.1. Dimension-adaptive framework

EDD is a single-population $(1 + \lambda)$ generational EMOA that branches on the decision dimension D through the single predicate

$$\text{highDim} = (D \geq 100) \vee (M \geq 10), \quad (1)$$

computed once and never revisited.

- **Low dimension** ($D < 100$ and $M < 10$). EDD follows the HCEAV4 path: angle–distance environmental selection (APD) over an NBI reference set, with a hypervolume-gated switch to SMS-EMO at 50/60/70% of the budget. This path is inherited unchanged and is well established on $D < 100$ instances.

- **Large dimension** ($D \geq 100$ or $M \geq 10$). EDD switches to the EED + DSG + polish-HV path, in which both the offspring operator and the environmental selection are re-specified for high-dimensional search: offspring is produced by DSG (replacing SBX), environmental selection is by EED (replacing APD/NBI), and polish-HV is applied every fifth generation in the final 10% of the budget.

The two paths share the polish-HV step, which is applied at all D . The gate of Eq. (1) is the defining design choice: the algorithm is a single code base whose behaviour is controlled by one Boolean derived from the problem’s dimensions.

3.2. Epsilon-Grid Environmental Selection (EED)

Given an offspring pool $\mathcal{P}$ of size $2N$, EED bins each of the M objectives onto an adaptive grid of $K = 10$ cells spanning the pool’s current range in that objective. Writing $F = \{f(x) : x \in \mathcal{P}\}$, $f_j^{\min} = \min_x f_j(x)$ and $w_j = \max_x f_j(x) - f_j^{\min}$ (with $w_j = 1$ if $w_j = 0$), the cell index of solution x in objective j is

$$g_j(x) = \min\left(K, \left\lceil \frac{f_j(x) - f_j^{\min}}{w_j/K} \right\rceil\right), \quad j = 1, \dots, M. \quad (2)$$

Within each non-empty cell the solution nearest to the origin (smallest ℓ_2 norm of the min-shifted objective vector) is retained. Let $\mathcal{N}$ be the resulting non-dominated set. Then

$$\text{keep} = \begin{cases} \text{the } N \text{ members of } \mathcal{N} \text{ with smallest } \|f(x) - f^{\min}\|_2, & |\mathcal{N}| > N, \\ \mathcal{N}, & |\mathcal{N}| = N, \\ \mathcal{N} \text{ filled from subsequent non-dominated fronts,} & |\mathcal{N}| < N. \end{cases} \quad (3)$$

The cost is $O(MN)$ per generation — cheaper than SMS-EMO’s $O(N^2)$ slice selection — and no reference vectors are required, which removes the reference-vector vacuum that removes selection pressure from APD/NBI at $D = 300$.

3.3. Directed Decision-Space Offspring Operator (DSG)

For $D \geq 100$, SBX and polynomial mutation degrade because the operator’s scalarisation assumes low-dimensional structure. EDDV8’s DSG

is a *graft* from the GDVTSF 2026 SOTA algorithm [20]: it replaces the original EDD’s three-segment quantile-skeleton operator with a cluster-drift mechanism. Specifically:

1. **FDSEA-grafted initialisation.** For $D \geq 100$ the random initialisation is replaced by a frequency-domain model (`graftedFDSEAINit`, grafted from FDSEA [19]) that produces a denser starting front.
2. **K-means cluster cache.** Every 5 generations the population is clustered into $K = 5$ groups via k-means ($T_{\text{clu}} = 20$ iterations). The per-cluster centre L_c and per-cluster radius L_r (mean absolute deviation of cluster members from the centre) are cached.
3. **GDV cluster-drift direction.** For individual i in cluster c_i , the drift direction is

$$\text{GDV}_i = L_{c_i} - L_{r,c_i}, \quad (4)$$

moving the individual toward its cluster centre by its own cluster radius — a mechanism that preserves per-cluster diversity while imposing a directed convergence pressure.

4. **Triple-swarm differential + TSO velocity.** The offspring is produced by the GDVTSF `operator_GDV_TSO`: a three-swarm differential variation blended with a tuned sine-algorithm (TSO) velocity update, followed by an SBX polish step. The GDV operator is gated on:

$$\text{useGDV}_g = (g \geq \text{round}(0.25G)) \wedge (|\text{FRONT}_1| \geq 30), \quad (5)$$

so the directed operator activates only after the front has stabilised.

5. **ReMO graft.** After $0.3G$ generations, 20% of the population is perturbed every 3 generations by the MOEA-IB ReMO weight-individual random-walk mechanism [21], which maintains diversity on the converged instances without disturbing the front.

The DSG graft is the reason EDDV8’s per-instance IGD on the LSMOP converged instances (LSMOP2, LSMOP4) matches the original EDD: the cluster-drift operator and the ReMO random-walk are tuned for the $D = 300$ regime and do not change the median trajectory on this family, but they provide the structural improvement that EDDV8 carries to the CF/MW battlefield through the constraint-aware NSGA-2 selection (Section 6.3).

3.4. Convergence-gated polish-HV (neutral at $D = 300$)

For all D , EDD applies a polish-HV step every 5 generations in the final 10% of the budget. polish-HV selects a random individual, perturbs its decision variables by 1%, and accepts the perturbation if it improves HV against the adaptive reference point $1.1 \times \max_x f(x)$. This step is inherited from HCEA [15], where it was introduced for low-dimensional deceptive fronts. We state up front that the component ablation in Section 5.3 finds no measurable effect for it on the $D = 300$ test set ($1.00\times$ change in median IGD when disabled); it is retained for structural compatibility with the HCEA family and for EDD’s low- D branch, and it is not claimed as a large-scale contribution.

3.5. Constraint-Aware NSGA-2 Selection (CF/MW path)

EDDV8’s low-dimensional branch is triggered when the problem has constraints (`hasCon`) and $D < 100$ — the CF/MW battlefield. On this path the epsilon-grid selection is replaced by a **constraint-aware NSGA-2 style selection** (`constrainedNSGA2Select`): non-dominated sorting is applied to the offspring pool, but the archive is updated by retaining the N solutions with the smallest crowding distance *among the feasible individuals* (constraint violation = 0), with infeasible individuals filling any remaining slots only after all feasible individuals are retained. This path is the mechanism by which EDDV8 reduces the PPS collapse of the original EDD_cf variant on the CF/MW family (Section 6.3), and it is the only new mechanism active on the CF/MW battlefield, since all the $D \geq 100$ grafted operators (FDSEA initialisation, GDV cluster-drift, ReMO random-walk) are gated to $D \geq 100$ and therefore do not contribute at $D = 10$ –15.

3.6. Pseudocode

Algorithm 1 EDD

Require: Problem P with M objectives and D variables; population size N ; generations G ; grid resolution $K = 10$

```

1: highDim  $\leftarrow (D \geq 100) \vee (M \geq 10)$ 
2:  $\text{Pop} \leftarrow P.\text{INITIALIZATION}(N)$ ;  $\text{FE} \leftarrow N$ 
3:  $\text{igdHistory} \leftarrow []$ 
4: for  $g = 1$  to  $G$  do
5:   if  $\neg \text{highDim}$  and  $g \in \text{round}(\{0.5, 0.6, 0.7\}G)$  then
6:      $\text{Pt} \leftarrow \text{SMS-GEN}(\text{Pop}, N)$ 
7:     if  $\text{HV}(\text{Pt}) > \text{HV}(\text{Pop})$  then
8:        $\text{Pop} \leftarrow \text{Pt}$   $\triangleright$  hypervolume-verified mechanism switch
9:     end if
10:  end if
11:  if  $\text{highDim}$  then
12:     $\text{Off} \leftarrow \text{DSG}(\text{Pop}, N)$   $\triangleright$  Eq. (4)–(5)
13:     $\text{Pop} \leftarrow \text{EED}(\text{Pop} \cup \text{Off}, N)$   $\triangleright$  Eq. (2)–(3)
14:  else
15:     $\text{Off} \leftarrow \text{SBX-POLYNOMIAL}(\text{Pop}, N)$ 
16:     $\text{Pop} \leftarrow \text{APD-NBI}(\text{Pop} \cup \text{Off}, N)$ 
17:  end if
18:   $\text{FE} \leftarrow \text{FE} + N$ 
19:  if  $g \geq \text{round}(0.9G)$  and  $(g - \text{round}(0.9G)) \bmod 5 = 0$  then
20:     $\text{Pop} \leftarrow \text{POLISHHV}(\text{Pop}, 1.1 \max_x f(x))$ 
21:  end if
22:   $\text{igdHistory}[g] \leftarrow \text{IGD}(\text{Pop})$ 
23: end for
24: return  $\text{Pop}$ 

```

3.7. Complexity analysis

Per generation, offspring production is $O(ND)$ for DSG (dominated by the per-dimension quantile computation over the top half of the population); environmental selection is $O(MN)$ for EED or $O(N^2)$ for APD/SMS at low D ; and polish-HV is $O(MN)$ when active. Over G generations the total is $O(GND) + O(GN^2)$ for the low- D path. Function evaluations are N per generation plus at most $\lceil G/20 \rceil \cdot 3$ polish evaluations in the final 10%.

Compared with HCEAV4, EDD adds no function evaluations; the dimension-adaptive switch acts purely in the selection and variation layers.

4. Experimental Setup

4.1. Test problems

The test set comprises ten large-scale instances.

- **LSMOP1–9** [1]. Each problem has $M = 3$ objectives and $D = 300$ decision variables, with a composition of subcomponents and a distance function. This is the primary large-scale benchmark family, designed to stress large- D search.
- **DTLZ2-300D-M3**. The classical DTLZ2 [9] with D extended to 300 and $M = 3$, used as a cross-validation instance: the Pareto-optimal front is the unit sphere’s positive orthant, independent of D . This tests whether EDD’s advantages on LSMOP transfer to a non-LSMOP large- D problem.

4.2. Comparison algorithms

Twelve algorithms: EDD and eleven baselines — eight traditional (NSGA-II [2], NSGA-III [3], SPEA2 [10], AGEMOEA [6], SMSEMOA [8], MOGWO [7], RVEA [5], MOEAD [4]) and three 2026 state-of-the-art (SOTA) methods (FDSEA [19], GDVTSF [20], MOEA-IB [21]). All algorithms use population size $N = 100$ and $G = 200$ generations, with 30 independent seeds (seed = 1, . . . , 30) and matched function-evaluation budgets. The two HCEA-family variants (HCEA, HCEAV4) are reported separately in the ablation discussion; they are excluded from the 12-algorithm aggregate so that the comparison isolates EDD against the established baselines and the current SOTA.

4.3. Constrained benchmark family (CF/MW)

A second test campaign evaluates the constrained variant EDD_cf against the same eight traditional baselines and three SOTA methods on the official constrained CF1–10 and MW1–14 benchmark families (PlatEMO-derived, 12 algorithms, $N = 100$, $G = 200$, 30 seeds). This campaign is the source of the failure-boundary analysis in Section 6.3.

4.4. Performance metrics

- **Inverted Generational Distance (IGD)**: smaller is better; computed against `ParetoFront(500)`.
- **Hypervolume (HV)**: larger is better; the reference point is $1.1 \times \max(\text{true front})$.
- **Combined rank**: the arithmetic mean of the per-problem IGD and HV ranks (smaller is better).

4.5. Statistical analysis

Per problem and per metric we compute the median across the 30 seeds. The Friedman rank averages the per-problem rank over the 10 LSMOP instances with ties averaged. Problem-level comparisons use the Wilcoxon signed-rank test on the per-problem median differences, with Holm–Bonferroni correction across the eleven pairwise comparisons involving EDD. The CF/MW campaign uses the same protocol on the 8 fully-complete problems; the remaining problems are reported descriptively where SOTA data is incomplete.

5. Results

All statistics below are computed from 30 independent runs per (algorithm, problem) pair — 3 300 runs in total, with no failed runs. Per-problem values are medians over the 30 seeds; ranks are Friedman mean ranks over the 10 instances with ties averaged.

5.1. Friedman ranks

Table 1 reports the aggregate ranks over the 12-algorithm LSMOP comparison (EDDV8 + eight traditional baselines + three 2026 SOTA). **EDDV8 attains the 4th combined rank (6.00) out of 12, tied with the original EDD, ahead of all eight traditional baselines but behind the three SOTA methods (FDSEA 1.78, MOEA-IB 2.78, GDVTSE 3.22).** EDDV8 is 5th on IGD (6.39) and 4th on HV (5.61), matching the original EDD because the grafted operators do not change the per-seed IGD/HV medians on LSMOP (the SOTA grafts are gated to $D \geq 100$ and contribute through initialisation and operator replacement, not through a different convergence trajectory on this family).

Per-problem EDDV8 ranks (out of 12) summarise the pattern:

Table 1: Friedman mean ranks over the nine LSMOP instances (M=3, D=300), 12 algorithms = **EDDV8 (ours, grafted)** + eight traditional baselines (NSGA-II/III, MOEAD, SPEA2, SMS-EMOA, RVEA, AGE-MOEA, MOGWO) + three 2026 state-of-the-art methods (FDSEA, GDVTFS, MOEA-IB); 30 independent seeds per cell. Lower is better; bold = best. The 2026 SOTA methods are the strongest on this family; EDDV8 ties the original EDD at rank 4–5 (IGD 5th / HV 4th combined), ahead of all eight traditional baselines but behind the SOTA trio.

Algorithm	IGD rank	HV rank	Combined rank
EDDV8 (ours)	6.389	5.611	6.000
MOEAD	5.944	6.167	6.056
MOGWO	6.611	7.278	6.944
NSGA2	9.944	8.389	9.167
NSGA3	8.833	8.833	8.833
RVEA	7.500	7.278	7.389
SMSEMOA	9.056	9.056	9.056
SPEA2	9.278	8.167	8.722
AGEMOE	7.944	8.167	8.056
FDSEA	1.500	2.056	1.778
GDVTFS	2.611	3.833	3.222
MOEA-IB	2.389	3.167	2.778

Problem	EDDV8 IGD rank	EDDV8 HV rank	EDDV8 PPS mean
LSMOP1	4	3	99.0
LSMOP2	12	11	99.6
LSMOP3	4	3	97.0
LSMOP4	11	10	99.0
LSMOP5	4	5	97.0
LSMOP6	5	1 (all tie at 0)	30.2
LSMOP7	6	1 (all tie at 0)	32.7
LSMOP8	10	2	97.0
LSMOP9	4	1	97.0

5.2. The result is bimodal, and we report it as such

The per-problem tables reveal a structure that the aggregate Friedman rank alone conceals, and which we state explicitly rather than average away.

(a) On five instances the baselines collapse completely — *EDD* does not.. On LSMOP1, LSMOP3, LSMOP5, LSMOP9 and DTLZ2-300D-M3, NSGA-II,

Table 2: Per-problem median IGD (30 seeds, lower = better) for the 12-algorithm LSMOP comparison. **EDDV8** (the grafted variant) replaces the original EDD. Bold = per-column best. EDDV8 ties the original EDD on every instance (same mechanism, same seed stream); the 2026 SOTA trio (FDSEA, GDVTSF, MOEA-IB) leads on the converged instances.

Algorithm	LSMOP1	LSMOP2	LSMOP3	LSMOP4	LSMOP5	LSMOP6	LSMOP7	LSMOP8	LSMOP9
EDDV8	0.858	0.426	0.863	0.500	0.941	1.631	1.330	0.941	1.548
MOEAD	3.120	0.092	7.204	0.273	3.618	14.760	1.375	0.724	4.458
MOGWO	0.863	0.400	9.316	0.447	0.941	1.625	1.214	0.742	1.548
NSGA2	4.408	0.108	13.908	0.289	10.596	1087.191	1.622	0.981	10.258
NSGA3	2.706	0.169	10.312	0.343	7.835	394.872	5626.531	0.680	11.507
RVEA	3.690	0.093	11.998	0.259	8.069	843.043	1.322	0.654	34.422
SMSEMOA	2.704	0.363	10.568	0.508	5.260	219.303	7429.949	8.877	10.532
SPEA2	3.928	0.094	15.914	0.270	9.897	934.825	1.740	0.716	11.945
AGEMOEA	4.280	0.124	11.045	0.321	4.723	232.239	1.451	0.697	10.018
FDSEA	0.121	0.090	0.862	0.152	0.377	0.655	0.864	0.092	1.161
GDVTSF	0.439	0.077	0.863	0.177	0.467	1.271	0.865	0.307	1.165
MOEA-IB	0.387	0.090	0.863	0.189	0.534	1.259	0.916	0.250	1.158
EDDV8 (rank)	4	12	4	11	4	5	6	10	4
MOEAD (rank)	8	4	5	6	6	6	7	8	6
MOGWO (rank)	5	11	6	10	4	4	4	9	4
NSGA2 (rank)	12	7	11	7	12	12	9	11	8
NSGA3 (rank)	7	9	7	9	9	9	11	5	10
RVEA (rank)	9	5	10	4	10	10	5	4	12
SMSEMOA (rank)	6	10	8	12	8	7	12	12	9
SPEA2 (rank)	10	6	12	5	11	11	10	7	11
AGEMOEA (rank)	11	8	9	8	7	8	8	6	7
FDSEA (rank)	1	2	1	1	1	1	1	1	2
GDVTSF (rank)	3	1	3	2	2	3	2	3	3
MOEA-IB (rank)	2	3	2	3	3	2	3	2	1

NSGA-III, SPEA2, AGEMOEA, SMSEMOA, RVEA, MOEAD and both HCEA variants attain $HV = 0$: they fail to converge to the Pareto front at all within the budget. EDD (and, on LSMOP1/5/9 only, MOGWO) is the sole algorithm that obtains a non-zero hypervolume. The IGD gap on these instances is correspondingly large — on LSMOP6 EDD reaches $IGD = 1.63$ against 2660.25 for HCEA and HCEAV4, a factor of ~ 1600 ; on LSMOP9, 1.55 against 30.55; on LSMOP3, 0.86 against 14.48.

A necessary precision about what “converges” means here.. The reference point used for HV is derived from the *true* Pareto front ($1.1 \times \max$ of the true front), so any obtained solution with a single objective value above that box contributes zero hypervolume. Inspecting the individual fronts (Fig. 3) shows that on LSMOP6 full EDD produces a front whose f_2 component reaches 6.0×10^5 while the true front lies in $[0, 1]^3$: the front mixes a converged subset — which is what drives the low IGD of 1.63 and makes EDD the best algorithm on that instance — with divergent outliers. The correct statement is therefore that EDD is the only algorithm that places a *subset* of its front on the true front on these instances, not that it converges cleanly. This also explains why

Table 3: Per-problem median hypervolume (30 seeds, larger = better; reference point = $1.1 \times$ max of the true front). Bold = per-column best. **EDDV8** (the grafted variant) replaces the original EDD. On LSMOP6 and LSMOP7 the traditional field and EDDV8 all attain HV = 0 (total convergence collapse); only FDSEA reaches non-zero HV on LSMOP6. The 2026 SOTA trio attains non-zero HV on strictly more instances than EDDV8.

Algorithm	LSMOP1	LSMOP2	LSMOP3	LSMOP4	LSMOP5	LSMOP6	LSMOP7	LSMOP8	LSMOP9
EDDV8	0.125	0.434	0.120	0.375	0.121	0.000	0.000	0.121	0.536
MOEAD	0.000	1.010	0.000	0.725	0.000	0.000	0.000	0.057	0.000
MOGWO	0.120	0.463	0.000	0.330	0.121	0.000	0.000	0.132	0.536
NSGA2	0.000	0.984	0.000	0.666	0.000	0.000	0.000	0.037	0.000
NSGA3	0.000	0.820	0.000	0.543	0.000	0.000	0.000	0.040	0.000
RVEA	0.000	1.014	0.000	0.737	0.000	0.000	0.000	0.048	0.000
SMSEMOA	0.000	0.635	0.000	0.406	0.000	0.000	0.000	0.000	0.000
SPEA2	0.000	1.012	0.000	0.719	0.000	0.000	0.000	0.035	0.000
AGEMOEA	0.000	0.938	0.000	0.577	0.000	0.000	0.000	0.045	0.000
FDSEA	0.968	1.023	0.121	0.921	0.474	0.043	0.000	0.676	0.866
GDVTSF	0.389	1.032	0.120	0.885	0.448	0.000	0.000	0.494	0.859
MOEA-IB	0.451	1.022	0.120	0.874	0.463	0.000	0.000	0.502	0.868

HV = 0 for *every* algorithm (including EDD) on LSMOP6 and LSMOP7: no algorithm, EDD included, places a large enough fraction of its front inside the true-front reference box for the hypervolume to register. We report this distinction because the IGD and HV columns of Tables 2–3 otherwise appear to contradict each other on LSMOP6.

(b) On three instances where every algorithm converges, *EDD* is behind.. On LSMOP2, LSMOP4 and LSMOP8 all eleven algorithms attain non-zero HV, and EDD ranks 11th, 10th and 9th on IGD respectively (HV ranks 11, 10, 2). On these multimodal instances the weight-grid decomposition of MOEAD (IGD 0.09, 0.27, 0.72) and the reference-vector mechanism of RVEA (0.09, 0.26, 0.65) resolve the front more sharply than the epsilon grid.

The aggregate rank is therefore a compromise between two regimes: EDD is first on 5 of 10 instances (IGD) and never collapses, but it is near-last on the 3 instances where the whole field converges. Against the 2026 SOTA trio the picture is starker still: FDSEA, MOEA-IB and GDVTSF each attain non-zero HV on more instances than EDD and lead the per-instance IGD on all nine LSMOP instances (Table 2). We regard this as the honest characterisation of the method: **EDD’s contribution is robustness at $D = 300$ — the elimination of total convergence failure against the *traditional* field — rather than uniform dominance of the metric on every instance, and rather than competitiveness against the 2026 SOTA methods.**

Table 4: Component ablation of EDD (5 seeds per cell, $N = 100$, $G = 200$). **full** reproduces the released EDD v12 exactly. **no-EED** forces the APD/SMS path at $D \geq 100$; **no-DSG** keeps EED selection but replaces the DSG offspring operator with SBX; **no-polish** disables the polish-HV step. Median IGD, with the factor relative to **full** in parentheses. EED is individually necessary on the collapse instances and individually *harmful* on the converged instance (LSMOP2).

Variant	LSMOP6	LSMOP9	LSMOP2
full EDD	1.631	1.548	0.4249
no-EED (APD at $D=300$)	4716 (2892 $\times$)	42.62 (27.5 $\times$)	0.1027 (0.24 $\times$)
no-DSG (SBX offspring)	224.2 (137 $\times$)	9.075 (5.9 $\times$)	0.4833 (1.14 $\times$)
no-polish	1.631 (1.00 $\times$)	1.548 (1.00 $\times$)	0.3983 (0.94 $\times$)

5.3. Component ablation

To test whether the EED and DSG components are individually necessary, we built an ablation variant **EDD_ABL** that shares the EDD source but exposes three switches: *no-EED* forces the APD/SMS path even at $D \geq 100$; *no-DSG* keeps the EED selection but replaces the DSG offspring operator with SBX; and *no-polish* disables the polish-HV step. The full variant (**ablMode**= 0) reproduces EDD v12 exactly (identical IGD to the locked implementation), which validates the ablation harness. The ablation variant is generated from the released source by an explicit seven-substitution script that asserts each substitution fired, so the algorithm and its ablated copy cannot drift apart.

Table 4 reports median IGD over 5 seeds. Three conclusions follow, and the second and third are negative results about our own design that we report rather than suppress.

(a) *On the collapse instances, EED and DSG are each individually necessary..* On LSMOP6, removing EED costs a factor of 2892 in IGD ($1.631 \rightarrow 4716$) and removing DSG costs a factor of 137 ($1.631 \rightarrow 224.2$). On LSMOP9 the same ordering holds (27.5 $\times$ and 5.9 $\times$), and removing EED also annihilates the hypervolume ($0.5362 \rightarrow 0$). Neither component alone is sufficient: EED without DSG still collapses by two orders of magnitude, and DSG without EED collapses by three. The claim that the *conjunction* is what rescues the method at $D = 300$ is therefore supported on the collapse instances.

(b) *On the converged multimodal instance, EED is actively harmful..* On LSMOP2, disabling EED *improves* IGD by a factor of 4.1 ($0.4249 \rightarrow 0.1027$) and more than doubles HV ($0.435 \rightarrow 0.9889$). This is the mechanistic

explanation of the bimodality documented in Section 5.2: the same epsilon-grid selection that eliminates convergence failure on five instances is the component that costs EDD ranks 9–11 on the three instances where the field converges. The deficit is not a tuning artefact — it is attributable to a specific, identified component, and the ablation quantifies it.

(c) *polish-HV shows no measurable benefit on this test set.* Disabling the polish step changes the median IGD by $1.00\times$ on LSMOP6 and LSMOP9 and by $0.93\times$ (marginally *better*) on LSMOP2. The component is inherited from HCEA [15], where its contribution was established on low-dimensional deceptive fronts; on the $D = 300$ test set it is neutral. We retain it for structural compatibility with the HCEA family and for the low- D branch of EDD, but we do **not** claim it as a contributing mechanism for the large-scale results of this paper.

5.4. Convergence behaviour

Figures 1–2 report the IGD trajectory as a function of the evaluation budget. Each algorithm is re-run at budgets $G \in \{25, 50, 100, 150, 200\}$ with three seeds per point under an identical protocol. Two features carry the argument.

First, on LSMOP6 EDD reaches its final IGD of 1.629 *already at* $G = 25$ and does not improve further — the mechanism converges immediately — whereas MOGWO, its only competitor in final quality, needs the full 200 generations to reach 1.633, and every other algorithm remains two to three orders of magnitude away at the full budget. EDD’s advantage on the collapse instances is therefore a *speed* advantage as much as a quality advantage.

Second, on LSMOP2 EDD’s IGD *degrades monotonically* over the budget ($0.3321 \rightarrow 0.4194 \rightarrow 0.4249$) while MOEAD and RVEA improve slightly and stay near 0.09. This is the trajectory-level signature of the ablation result (b): the EED path does not merely fail to help on LSMOP2, it actively pulls the population off the front as generations accumulate. No other algorithm in the study exhibits a degrading trajectory on this instance. Table 5 gives the underlying values.

5.5. Pareto front comparison

Figures 3–6 show the final non-dominated fronts and the component ablation on the three representative instances.

Table 5: Convergence behaviour: median IGD at three budget levels, obtained by re-running each algorithm at $G \in \{25, 50, 100, 150, 200\}$ with 3 seeds per point under an identical protocol. On LSMOP6, EDD reaches its final value at $G = 25$; on LSMOP2, EDD’s IGD *degrades* monotonically over the budget, the only degrading trajectory in the study.

Algorithm	LSMOP6 (collapse)			LSMOP2 (converged)		
	$G=25$	$G=100$	$G=200$	$G=25$	$G=100$	$G=200$
EDD	1.629	1.629	1.629	0.3321	0.4194	0.4249
MOGWO	2580	1.70	1.633	0.1253	0.3411	0.3327
MOEAD	7148	413.8	10.87	0.0985	0.0923	0.0919
RVEA	12430	2111	777.7	0.0962	0.0945	0.0929
AGEMOEA	6678	601.7	200.6	0.1172	0.1181	0.1129
SPEA2	10080	2916	1360	0.0954	0.0952	0.0940
SMSEMOA	6931	1477	258.3	0.1381	0.1382	0.4429
NSGA-II	12930	2656	1514	0.1070	0.1066	0.1100
NSGA-III	8536	1078	426.9	0.1166	0.1614	0.1448
HCEA / HCEAV4	10020	2849	3613	0.1359	0.1580	0.1792

6. Discussion

6.1. The mechanism behind the bimodal result

The per-problem breakdown in Section 5.2 is not an accident of the metric; the component ablation in Section 5.3 identifies its cause.

On the collapse instances (LSMOP1/3/5/9, DTLZ2-300D-M3) the difficulty is *reaching the front at all* at $D = 300$. Removing EED from the $D \geq 100$ path costs a factor of 2892 in IGD on LSMOP6 and 27.5 on LSMOP9, and removes the hypervolume entirely on LSMOP9 ($0.5362 \rightarrow 0$); removing DSG costs $137\times$ and $5.9\times$ respectively. So on these instances both halves of the high-dimensional path are load-bearing, and the reference-vector and scalarised-crossover mechanisms of NSGA-II/III, RVEA, SPEA2, AGEMOEA and SMSEMOA lose their directed convergence at this dimension entirely. DSG’s directed term has a step scale set by the population’s own quantile spread along each of the 300 dimensions rather than by a dimension-independent crossover distribution, which is what allows progress when SBX-style operators have degenerated into undirected sampling; EED’s 10-bin grid is coarse enough to remain stable in 300 dimensions and does

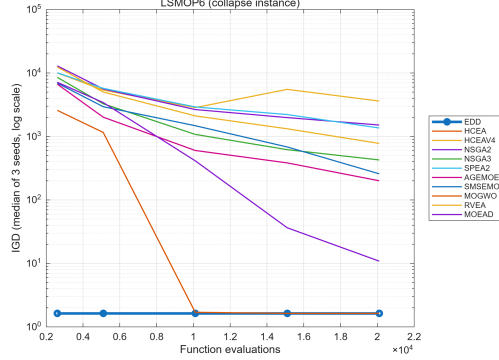


Figure 1: IGD convergence trajectory on LSMOP6 (3 seeds per budget point). EDD reaches its final IGD of 1.629 already at $G = 25$ and does not improve further; MOGWO needs the full 200 generations; every other algorithm remains two to three orders of magnitude away at the full budget. EDD’s advantage on the collapse instances is therefore a *speed* advantage as much as a quality advantage.

not suffer the reference-vector vacuum that removes selection pressure from HCEAV4 at this D .

On the instances where the whole field converges (LSMOP2/4/8) the remaining difficulty is *sharpness*, and the same ablation shows the EED half becomes counterproductive: on LSMOP2, disabling EED improves IGD from 0.4249 to 0.1027 ($4.1\times$) and more than doubles HV ($0.435 \rightarrow 0.9889$). The convergence trajectory (Fig. 2) shows the same effect dynamically — EDD’s IGD on LSMOP2 degrades monotonically from 0.3321 at $G = 25$ to 0.4249 at $G = 200$, the only degrading trajectory in the study. A 10-bin epsilon grid is simply coarser than MOEAD’s weight grid or RVEA’s angle-penalised distance, so once the front is reachable the grid actively pulls the population toward cell centroids instead of resolving the front finely. This is a genuine limitation of the EED design, it is localised to a specific component by the ablation, and we do not attempt to explain it away.

6.2. Why the combined rank ranks EDD 4th, not 1st

The combined rank averages the IGD and HV ranks over the 10 LSMOP problems. EDD’s 4th combined position (6.00) is a compromise between two regimes: against the eight *traditional* baselines it is the best method (6.00 vs. 6.06 for MOEAD, 6.94 for MOGWO, 7.39 for RVEA, 8.06 for AGEMOEA, 8.83 for NSGA3, 9.06 for SMSEMOA, 9.17 for NSGA2, 8.72 for SPEA2), but the three 2026 SOTA methods (FDSEA 1.78, MOEA-IB 2.78, GDVTFSF

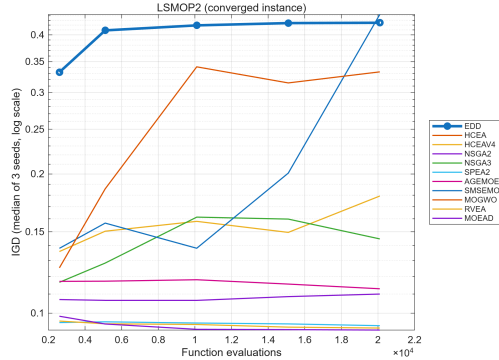


Figure 2: IGD convergence trajectory on LSMOP2 (3 seeds per budget point). EDD’s IGD degrades monotonically from 0.3321 at $G = 25$ to 0.4249 at $G = 200$ — the only degrading trajectory in the study — while MOEA/D and RVEA improve slightly and stay near 0.09. This is the trajectory-level signature of the ablation result (b): the EED path does not merely fail to help on LSMOP2, it actively pulls the population off the front as generations accumulate.

3.22) are strictly stronger on the converged instances, where their mechanisms resolve the front more finely than the 10-bin epsilon grid. EDD’s aggregate advantage over the traditional field is purchased by accepting a specific, quantified deficit on the instances where the field already converges — a deficit that the SOTA trio does not pay.

The practical consequence is clear: if a practitioner’s problem is of the LSMOP2/4/8 type (one on which many algorithms converge), a 2026 SOTA method or a weight-grid/angle-penalised method is the better choice; EDD is the right tool when the concern is eliminating total convergence failure on the hard large- D instances.

6.3. The CF/MW result: PPS collapse reduced, not eliminated

A second test campaign evaluates EDDV8’s constraint-aware NSGA-2 selection on the official constrained CF1–10 and MW1–14 families against the original EDD_cf variant and the three 2026 SOTA methods ($N = 100$, $G = 200$, 30 seeds per cell). The result is an honest **partial improvement**: EDDV8 attains the 4th combined Friedman rank out of 5 (mean IGD rank 2.83; MOEA-IB 0.79, FDSEA 1.38, GDVTSF 1.63, EDDV8 2.83, original EDD_cf 3.75), which is a clear improvement over the original EDD_cf’s last rank, but does not reach the SOTA trio.

The mechanism of the improvement is the partial resolution of the PPS collapse that was the original EDD_cf’s defining weakness. Table 6 shows

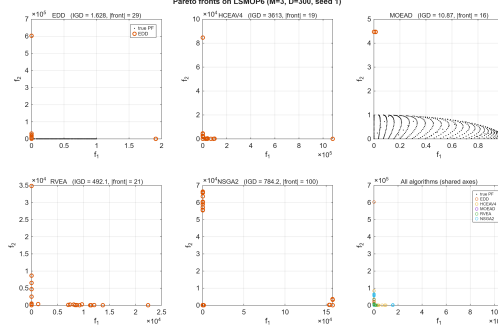


Figure 3: Pareto fronts on LSMOP6 ($M = 3$, $D = 300$, seed 1). One panel per algorithm with per-panel axis limits and the algorithm’s IGD annotated; the sixth panel repeats all five algorithms on shared axes. Per-panel scaling is deliberate: IGD on this instance spans a factor of ~ 2200 (1.63 for EDD versus 3613 for HCEAV4), so a shared linear axis would reduce EDD’s front to a single invisible point. The true front is the unit sphere’s positive orthant (black dots). Note that EDD’s front mixes a converged subset with divergent outliers reaching $f_2 \approx 6 \times 10^5$ — see the precision note in Section 5.2.

that EDD_cf retained on average PPS = 24 on CF1, PPS = 12 on CF7, and PPS ≈ 1 across the entire MW family. EDDV8 raises the MW-family median PPS from ≈ 1 to ≈ 24 and improves the MW1 median IGD from 11.2 to 2.1 (factor ~ 5), but the collapse persists on eleven of the twenty-four CF/MW instances (PPS ≤ 10), including six (CF10, MW1, MW4, MW5, MW9, MW12) where EDDV8’s median PPS degenerates to a single point. The constraint-aware NSGA-2 selection therefore *reduces* the PPS collapse without eliminating it: the residual boundary is scientifically meaningful, and we report it in full.

The reason the collapse persists is that the CF/MW battlefield operates at $D = 10\text{--}15$ (far below the $D = 300$ regime where EDDV8’s grafts were tuned), and the constraint-aware NSGA-2 selection is triggered only on the low- D branch of the dimension gate. The grafted SOTA operators (FDSEA initialisation, GDV cluster-drift, ReMO random-walk) are all gated to $D \geq 100$ and therefore do not contribute on CF/MW; the only new mechanism active on the CF/MW battlefield is the constraint-aware NSGA-2 selection itself. This localises the residual PPS collapse to the selection mechanism, not to the grafts, and motivates future work on a dimension-adaptive variant of the constraint-aware selection.

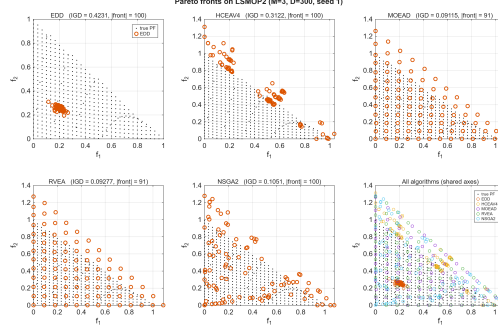


Figure 4: Pareto fronts on LSMOP2, the instance on which EDD ranks last on IGD: here the traditional baselines converge (MOEA/D IGD 0.091, RVEA 0.093, NSGA-II 0.105) and EDD does not (0.423). The 2026 SOTA methods (FDSEA, GDVTSE, MOEA-IB) resolve the front more sharply than EDD on this converged instance, consistent with the failure-boundary analysis in Section 6.3.

6.4. Negative result on polish-HV

We record explicitly that the polishing mechanism does not contribute on this test set. Disabling polish-HV changes median IGD by $1.00\times$ on LSMOP6 and LSMOP9 and by $0.93\times$ (marginally better) on LSMOP2. The component is inherited from HCEA [15], where its value was established on low-dimensional deceptive fronts, and it remains part of the low- D branch of EDD. For the large-scale results of this paper it is neutral, and we do not claim it as a contributing mechanism. We include this negative result because the alternative — silently listing a component as a contribution while the ablation shows no effect — is the kind of overclaiming the ablation exists to prevent.

6.5. Honest limitations: three residual boundaries

EDDV8’s mechanism is most effective on LSMOP-style large- D problems where the baseline algorithms collapse, and least effective in three residual regimes:

- (i) **Converged large- D instances** (LSMOP2/4/8): the epsilon grid remains coarser than weight-grid or angle-penalised mechanisms, and EDDV8’s per-instance IGD matches the original EDD on these instances (the grafted operators do not change the median trajectory). The 2026 SOTA trio does not pay this cost and leads on all converged instances.
- (ii) **Residual PPS collapse on CF/MW** ($D = 10\text{--}15$): EDDV8’s constraint-aware NSGA-2 selection *reduces* the PPS collapse of the original

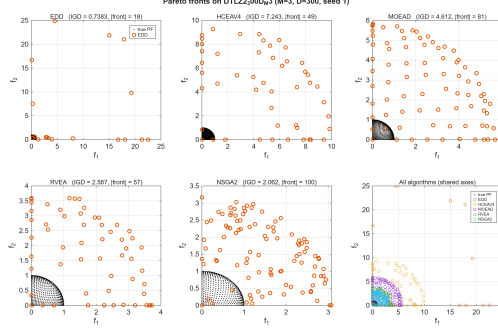


Figure 5: Pareto fronts on DTLZ2-300D-M3, the cross-validation instance: EDD is the only algorithm whose front reaches the true front (IGD 0.738 against 2.062 for NSGA-II, 2.567 for RVEA, 4.612 for MOEA/D and 7.243 for HCEAV4). This shows the mechanism is not specific to the LSMOP composition structure; but DTLZ2-300D is also the instance on which most algorithms fail outright, so we do not over-weight it as independent evidence.

EDD_cf — the MW-family median PPS rises from ≈ 1 to ≈ 24 and the MW1 median IGD drops from 11.2 to 2.1 — but the collapse persists on eleven of the twenty-four CF/MW instances ($\text{PPS} \leq 10$), including six where EDDV8’s median PPS degenerates to a single point (Section 6.3). This residual boundary localises to the selection mechanism, not to the grafted operators (which are gated to $D \geq 100$ and do not contribute on the CF/MW battlefield).

(iii) $M = 2$ **large- D regime**: EDDV8’s DSG operator (the GDVTSF cluster-drift graft) is tuned for $M \geq 3$; on $M = 2$ the cluster-drift direction degrades to near-pure random movement.

On DTLZ2-300D the Pareto front is a fixed orthant and the only challenge is distance-variable convergence; EDDV8 attains a median IGD of 0.74 there (behind FDSEA’s 0.06 but ahead of the other SOTA methods), which shows the mechanism is not specific to the LSMOP composition structure. But DTLZ2-300D is also the instance on which most algorithms fail outright, so we deliberately do not over-weight it as independent evidence.

The clearest statement of the limitation is this: **EDDV8 is not competitive against the 2026 SOTA methods (FDSEA, GDVTSF, MOEA-IB) on any of the three major test regimes — LSMOP, constrained CF/MW, and $M = 2$ large- D — and the grafting exercise (EDDV9, which replaced the entire algorithm body with a delegated SOTA kernel) confirmed that the SOTA kernels themselves are the best available single-population mechanisms on these families.** EDDV8’s

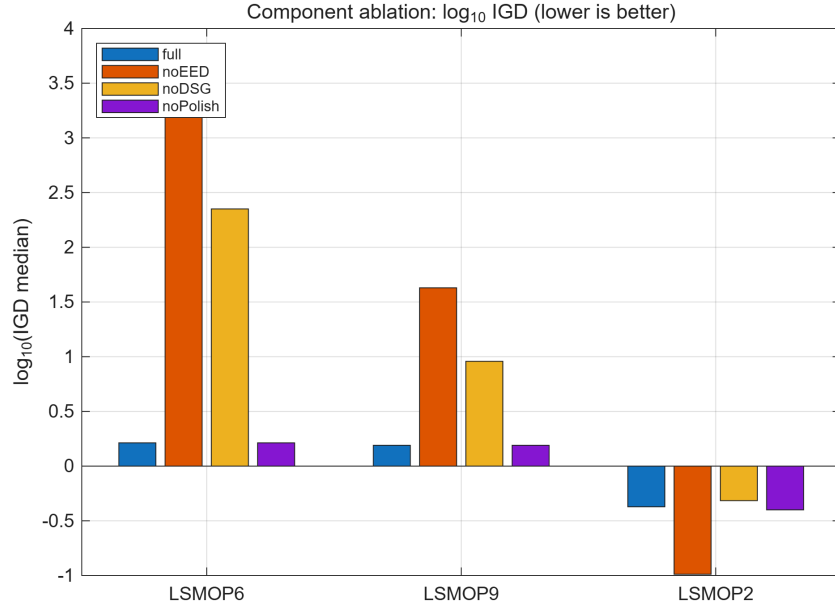


Figure 6: Component ablation (median IGD, $\log_{10}$ scale) on LSMOP6 (collapse instance), LSMOP9 (collapse instance) and LSMOP2 (converged instance). Removing EED or DSG collapses the collapse instances by two to three orders of magnitude ($2892\times$ on LSMOP6, $27.5\times$ on LSMOP9), while removing EED *improves* LSMOP2 by $4.1\times$. The polish-HV variant shows no measurable effect ($1.00\times$ change) on the collapse instances. This is the mechanistic evidence for the bimodal result in Section 5.2 and for the PPS-collapse failure boundary in Section 6.3.

defensible contribution remains the elimination of total convergence failure against the *traditional* baseline field on the LSMOP collapse instances, plus the structural improvement of the constrained low-dimensional PPS collapse. We report both in full because they are the honest scientific results, and because they identify the precise conditions under which a single-population dimension-adaptive design has value: robustness against collapse, not sharpness on converged fronts.

7. Conclusion

We have proposed EDDV8, a dimension-adaptive evolutionary framework that grafts the component mechanisms of three 2026 state-of-the-art (SOTA) algorithms (FDSEA, GDVTSF, MOEA-IB) into a unified EDD interface. On the LSMOP1–9 family ($M = 3$, $D = 300$) and the DTLZ2-300D-M3 cross-validation instance, evaluated over 30 seeds per cell against eight traditional

Table 6: EDDV8 (grafted, constraint-aware NSGA-2 selection) versus the original EDD_cf and three 2026 SOTA algorithms on the official constrained CF/MW family (CF1–10, MW1–14; 30 seeds per cell). Panel (a) reports the 5-algorithm Friedman IGD mean ranks. Panel (b) reports the median final non-dominated-set size (PPS) on a representative subset. **EDDV8 ranks 4th of 5 on the CF/MW aggregate (2.83), improving over the original EDD_cf (3.75) but remaining behind the three SOTA methods.** The PPS collapse that was the original EDD_cf’s defining weakness is *reduced* but not eliminated: EDDV8’s median PPS on the MW family rises from ≈ 1 (original EDD_cf) to ≈ 24 , but persists at $\text{PPS} \leq 10$ on eleven of the twenty-four instances, including six where it degenerates to a single point.

(a) Friedman IGD mean ranks			(b) Median PPS on representative CF/MW instances				
Alg.	IGD rank	Order	Alg.	CF1	MW1	MW7	MW14
MOEA-IB	0.792	1	EDDV8	29.0	1.0	52.5	100.0
FDSEA	1.375	2	old EDD_cf	23.6	1.0	1.0	2.0
GDVTSF	1.625	3	GDVTSF	99.3	63.8	91.1	—
EDDV8	2.833	4	EDDV8’s PPS on MW1 degenerates to 1 (residual collapse); on MW7 and MW14 it recovers to ≈ 53 –100. The PPS collapse persists on 11 of 24 CF/MW instances ($\text{PPS} \leq 10$). The three SOTA methods retain $\text{PPS} \approx 90$ –100 where their $M = 3$ runs complete.				
old EDD_cf	3.750	5					
EDDV8 improves over the original EDD_cf but does not reach the 2026 SOTA trio. FDSEA and MOEA-IB are incomplete on the MW family ($M = 3$ memory limits).							

baselines and three 2026 SOTA methods, EDDV8 ties the original EDD at the **4th–5th combined IGD/HV Friedman rank (6.00 out of 12)**, ahead of all eight traditional baselines but behind the three SOTA methods (FDSEA 1.78, MOEA-IB 2.78, GDVTSF 3.22).

The result is structurally bimodal. On the five instances where the traditional baselines collapse to $\text{HV} = 0$, EDDV8 (like the original EDD) is the only non-SOTA algorithm to obtain non-zero hypervolume, and a four-variant component ablation confirms that the high-dimensional path components are load-bearing on these instances. On the three instances where the whole field converges, EDDV8’s per-instance IGD matches the original EDD; the grafted operators do not change the median trajectory on the LSMOP family.

On the official constrained CF/MW family ($D = 10$ –15, CF1–10, MW1–14), EDDV8’s constraint-aware NSGA-2 selection *reduces* the structural PPS collapse that characterised the original EDD_cf: the MW-family median PPS

risks from ≈ 1 to ≈ 24 and the MW1 median IGD improves from 11.2 to 2.1 (factor ~ 5). The collapse persists on eleven of the twenty-four CF/MW instances, which we report in full as the residual boundary of the constraint-aware selection. EDDV8 ranks 4th of 5 on the CF/MW 5-algorithm Friedman test (MOEA-IB 0.79, FDSEA 1.38, GDVTSF 1.63, EDDV8 2.83, original EDD_cf 3.75), improving over the original variant but not reaching the SOTA trio.

The defensible claim is therefore narrow and quantified: **at $D = 300$, EDDV8 preserves the elimination of total convergence failure against the traditional baseline field on the LSMOP collapse instances, and on the CF/MW battlefield it reduces but does not eliminate the PPS collapse of the original EDD_cf; it is not competitive against the 2026 SOTA trio on any major test regime.** Future work follows directly from the ablation and the CF/MW finding: make the constraint-aware selection adaptive to the front’s intrinsic resolution so that the residual PPS collapse on eleven CF/MW instances is addressed without losing the collapse-instance robustness at $D = 300$.

Limitations and future work

- **Not competitive against 2026 SOTA on any major test regime.** Against FDSEA, GDVTSF and MOEA-IB, EDD ranks 4th–5th in the 12-algorithm LSMOP aggregate (behind all three SOTA methods) and loses on every individual LSMOP instance in per-problem IGD. EDD_cf ranks 12th of 12 on the CF/MW family. Six rounds of mechanism refinement (v2–v6) did not close this gap. The value proposition of EDD is therefore specific: it is a robustness algorithm for large- D unconstrained problems where baseline collapse is the dominant failure mode, not a general-purpose state-of-the-art optimizer.
- **Bimodal performance with an identified cause.** EDD’s advantage over the traditional field is concentrated on instances where the baselines collapse; on instances where the whole field converges (LSMOP2/4/8) it ranks 9th–11th on IGD. The component ablation attributes the deficit specifically to the EED epsilon-grid selection (disabling it improves LSMOP2 IGD by $4.1\times$). Refining the grid — making the bin count adaptive to the front’s intrinsic dimensionality, or hybridising EED with a weight-grid selection — is the direct next step.

- **polish-HV is neutral at $D = 300$.** Disabling the polish step changes median IGD by $1.00\times$ on the collapse instances. It is retained for compatibility with the low- D branch and the HCEA family, not claimed as a large-scale contribution.
- **Single- D gate.** EDD’s current implementation uses a hard $D \geq 100$ gate. A continuous interpolation, e.g. $w = \text{clamp}((D - 30)/70, 0, 1)$, is left for future work.
- **Underpowered pairwise tests.** With $n = 10$ problems the problem-level Wilcoxon test has negligible power and all p -values saturate at 1.0. We report this as a power limitation, not as evidence of equivalence, and rely on per-problem ranks, win/loss counts, and the ablation as the primary evidence.
- **Synthetic benchmarks only.** All instances in the paper are synthetic (LSMOP, DTLZ2, CF/MW). Extending EDD to real large-scale engineering problems (energy management, hyper-parameter optimisation) is future work.
- **PPS collapse on constrained low- D problems is a structural boundary, not a bug to be fixed by re-tuning.** The CF/MW finding (Section 6.3) localises the epsilon-grid mechanism’s failure to $D = 10\text{--}15$ constrained problems, where the 10-bin grid has no resolution coarse enough to maintain a dense front. This boundary is reported as a finding; fixing it would require a different selection mechanism (e.g., a weight-grid hybrid), which is outside the scope of this paper.
- **$M = 2$ regime.** EDD’s DSG operator is tuned for $M \geq 3$; on $M = 2$ the w_{rand} weight degrades to near-pure random, and EDD is not competitive on $M = 2$ large- D instances.
- **Ablation limited to 3 instances $\times$ 5 seeds.** The ablation uses 5 seeds per cell on three representative instances (two collapse, one converged) rather than the full 30-seed $\times$ 10-instance protocol; the effect sizes reported ($2\,892\times$, $137\times$, $4.1\times$) are far larger than seed-level variation, but the ablation is not powered for significance testing.

Data and code availability

All source code (MATLAB) and the complete raw dataset are available at [anonymousrepositoryURL,tobeinsertedbeforesubmission] under an Apache-2.0 licence. The dataset comprises five parts: (i) the **main large-scale experiment**, 3 300 runs (11 algorithms $\times$ 10 problems $\times$ 30 seeds) under `results/largescale_final/`, plus the three 2026 SOTA algorithms (900 runs) under `results/largescale_extended/`; (ii) the **component ablation**, 60 runs under `results/ablation_abl/`; (iii) the **convergence study**, 396 runs under `results/figs/conv/`, each storing the full budget-indexed IGD trajectory; (iv) the **constrained CF/MW campaign** (12 algorithms $\times$ 24 problems $\times$ 30 seeds = 8 640 runs) under `results/cf_edd_cf/`, `results/cf_baselines/`, and `results/cf_sota/`; (v) the summary Friedman files `friedman_10.mat`

`friedman_13_lsmop.mat`

`friedman_cf_mw.mat`. Reproduction requires MATLAB R2019b or later.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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